



MDMC

Master in Data Management
and Curation



Building Service Layers in the NFFA-DI Digital Ecosystem: Governed Bucket Management and Reusable Analysis Services

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Special thanks

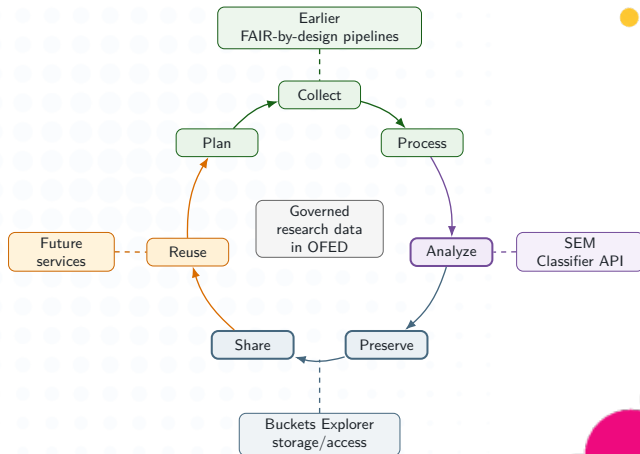
Data Engineering Laboratory
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Scholarships

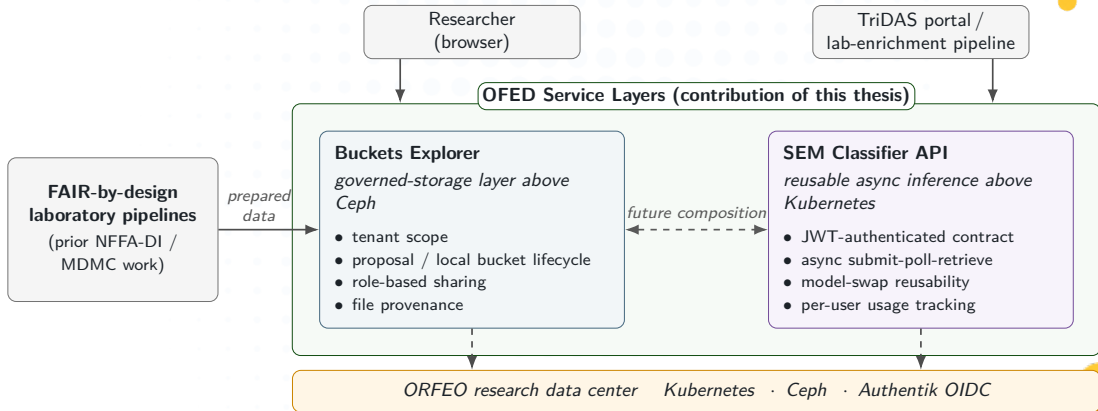
AREA Science Park: FAIR-
by-design track of the
MDMC program
SISSA: tuition fee
scholarship

The data-curation gap starts after acquisition

- OFED, the **Overarching FAIR Ecosystem for Data**, organizes NFFA-DI data after acquisition.
- Prepared data still need governed destinations, access policy, and provenance records.
- Analysis must become a callable service, not a model file or one-off script.



Thesis Overview

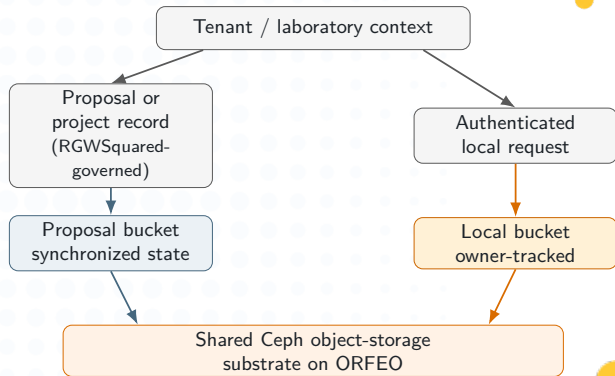


Buckets Explorer

governed object storage

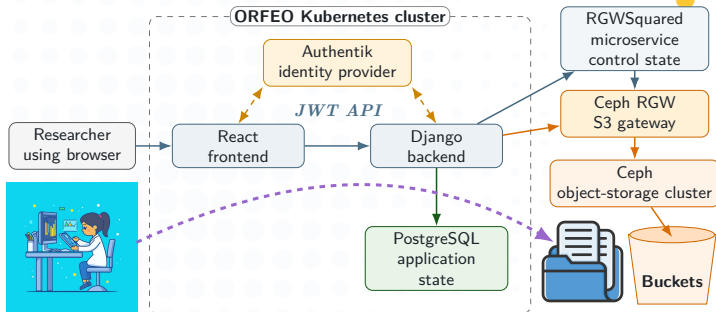
Why Buckets Explorer is needed

- Governed storage must stay inside facility infrastructure.
- Raw S3 lacks tenant, role, proposal, and provenance rules.
- Two lifecycles: proposal data and local work.

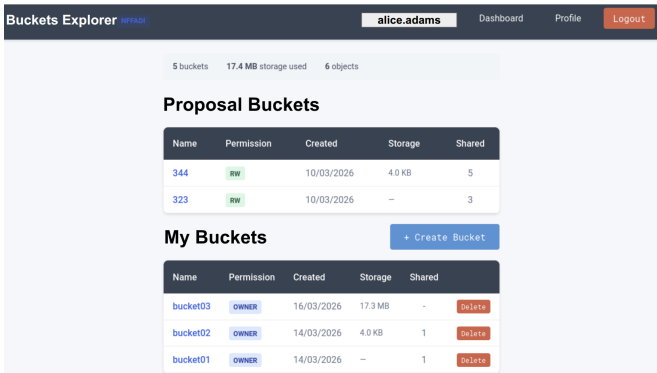


Method: split identity, control, data, and records

- **Identity** Authentik OIDC/JWT and tenant groups.
- **Control** Django policy checks; RGWSquared grants.
- **Data** Ceph RGW carries S3 traffic.
- **Records** PostgreSQL stores buckets, roles, uploads.



Result: a researcher-facing governed workspace



The screenshot shows the 'Buckets Explorer' interface for user 'alice.adams'. At the top, it displays '5 buckets', '17.4 MB storage used', and '6 objects'. Below this, there are two main sections: 'Proposal Buckets' and 'My Buckets'.

Proposal Buckets

Name	Permission	Created	Storage	Shared
344	RW	10/03/2026	4.0 KB	5
323	RW	10/03/2026	-	3

My Buckets

[+ Create Bucket](#)

Name	Permission	Created	Storage	Shared	
bucket03	OWNER	16/03/2026	17.3 MB	-	Delete
bucket02	OWNER	14/03/2026	4.0 KB	1	Delete
bucket01	OWNER	14/03/2026	-	1	Delete

- **Tenant gate** only ready tenants open.
- **Lifecycle** proposal buckets vs local buckets.
- **Roles** owner, RW, RO checked before grants.
- **Uploads** names and records preserve context.

Bucket and operator views

Buckets Explorer NFFAD alice.adams [Dashboard](#) [Profile](#) [Logout](#)

[← Back to Dashboard](#)

bucket02

OWNER Local Bucket

Share [Upload Files](#)

Owner: **alice.adams** · 2 users with access Hide

bob.baker	bob.baker@uo-code02.it	Read-Write
alice.adams	alice.adams@uo-code01.it	Owner

Name	Size	Uploaded By	Last Modified	Actions
nffadi-bucket02-uo-code01-filename05.pdf	1.5 KB	alice.adams	21/03/2026 18:50	Download View Delete
nffadi-bucket02-uo-code02-filename04.md	3.3 KB	bob.baker	14/03/2026 12:57	Download View Delete
nffadi-bucket02-uo-code01-filename03.nxs	78.1 KB	alice.adams	21/03/2026 18:50	Download View Delete
nffadi-bucket02-uo-code01-filename02.csv	196.6 KB	alice.adams	21/03/2026 18:50	Download View Delete
nffadi-bucket02-uo-code01-filename01.png	1002.2 KB	alice.adams	21/03/2026 18:50	Download View Delete

5 files

Researcher bucket view: objects, actions, and sharing context.

Bucket Manager NFFAD

Users

NAME	EMAIL	ROLE	STATUS	UO CODE	LAST LOGIN
bob baker	bob.baker@uo-code02.it	NFFAD	OK	uo-code02	11/03/2026
bob baker	bob.baker@uo-code02.it	NFFAD	OK	uo-code03	23/04/2026
charlie clark	charlie.clark@uo-code03.it	NFFAD	OK	uo-code01	11/03/2026
alice adams	alice.adams@uo-code01.it	NFFAD	OK	uo-code04	17/03/2026
alice adams	alice.adams@uo-code01.it	NFFAD	OK	uo-code04	17/03/2026
david davis	david.davis@uo-code04.it	NFFAD	OK	uo-code04	17/03/2026

Operator user view

Bucket Manager NFFAD

Buckets

NAME	STATUS	TYPE	STORAGE	UO CODE	UO CODE	UO CODE	UO CODE
420 NFFAD420	NFFAD	PROPOSAL	0.0	0	0	0	10/03/2026
4 users with access							
frank baker	frank.baker	frank.baker@uo-code05.it					OK
eve evans	eve.evans	eve.evans@uo-code06.it					OK
david davis	david.davis	david.davis@uo-code04.it					OK
charlie clark	charlie.clark	charlie.clark@uo-code03.it					OK
alice adams	alice.adams	alice.adams@uo-code01.it					OK
bob baker	bob.baker	bob.baker@uo-code02.it					OK
1 bucket02							
NFFAD1alice-adams-ov-it-it-bucket02	NFFAD	LOCAL	0.0	0	0	0	10/03/2026
2 users with access							
bob baker	bob.baker	bob.baker@uo-code02.it					OK
alice adams	alice.adams	alice.adams@uo-code01.it					OK

Operator bucket view

Storage FAIR is implemented as mechanisms

Findable

Institutional names
+ upload records.

Accessible

OIDC workspace +
S3 retrieval.

Interoperable

Ceph RGW/S3 +
open identity.

Reusable

Records support
accountable reuse.

**The contribution is governed storage
not storage capacity.**

SEM Classifier API

reusable asynchronous analysis

A model is not a reusable analysis service

- A model is not a facility service.
- Reuse lives in the async service machinery.
- New services change only the domain payload.
- Contract, queue, records, and deployment stay stable.

Wrong unit of reuse

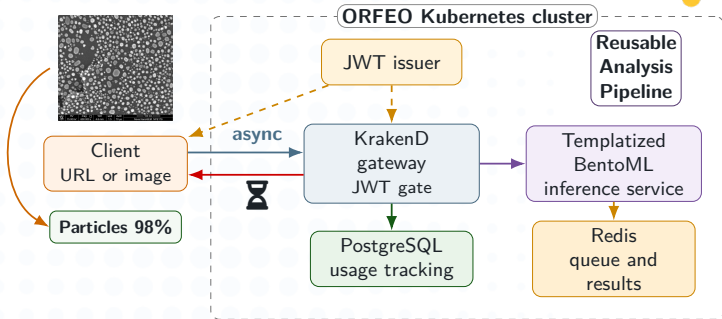
A model file plus ad hoc scripts.

Right unit of reuse

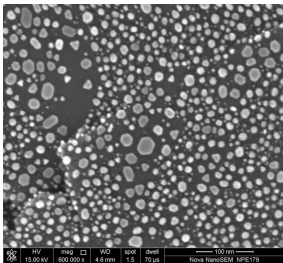
An authenticated async service template with an adaptable core.

Method: async submit, poll, retrieve

- **Gateway** JWT-checked public contract.
- **Input** URL or upload.
- **Queue** Redis job state and results.
- **Records** PostgreSQL request usage.



SEM Classifier API capabilities

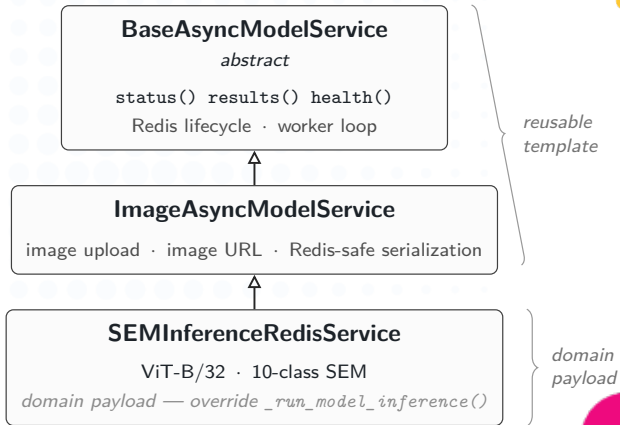


- **Contract** submit → job id → status → result.
- **Runtime** KrakenD, BentoML, Redis, PostgreSQL.
- **Payload** SEM-specific ViT-B/32 logic.
- **Accounting** user, endpoint, path, status.

Field	Returned value
Label	Particles
Confidence	97.98 %
Classes evaluated	10

Reusability comes from isolating the payload

- Stable: gateway, JWT, queue, results, records.
- Adaptable: model loading, inputs, inference.
- SEM classifier: one deployed instantiation.



Analysis FAIR is weaker, but still useful

Findable

Request records for operators.

Accessible

JWT REST: submit, status, result.

Interoperable

REST, OIDC/JWT, Kubernetes stack.

Reusable

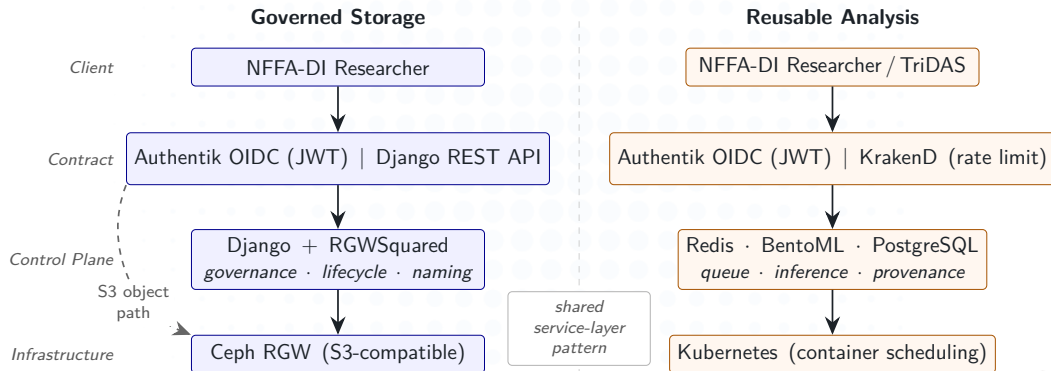
Same template, new payload.

Future storage integration could archive result objects; it is not deployed now.

Synthesis

The shared pattern is the transferable result

Independent implementations repeat the same responsibility split.



Limits and future work

Not claimed

No deployed end-to-end workflow currently sends Bucket Explorer objects directly into SEM analysis.

Not claimed

No generalized benchmark across facilities or storage-throughput claim.

Next step

Governed object URI → analysis job → governed result object.

Design rule

Compose services through contracts, authentication boundaries, records, and infrastructure data paths.

Conclusions

Governed storage

Buckets Explorer makes proposal and local bucket governance executable.

Reusable analysis

The SEM API exposes inference through an authenticated asynchronous contract.

Service-layer pattern

Public contract, auth boundary, control plane, data path, persistent record.

FAIR becomes credible when infrastructure enforces policy, access, execution, and records.

Open source, demos, and archived artifacts



 Buckets code



 SEM API code



 Live service



 Slides



 Buckets Explorer
Demo



Buckets Explorer Zenodo



Reusable Analysis Service Zenodo



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Italiadomani
PIANO NAZIONALE
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Thank you for your attention

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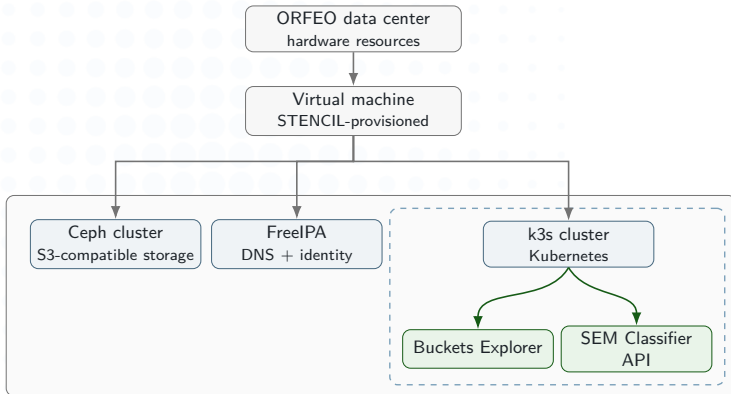
- “NFFA-DI cod. IR0000015, Missione 4, “Istruzione e Ricerca” – Componente 2, “Dalla ricerca all’impresa” – Linea di investimento 3.1, “Fondo per la realizzazione di un sistema integrato di infrastrutture di ricerca e innovazione” – Azione 3.1.1, “Creazione di nuove IR o potenziamento di quelle esistenti che concorrono agli obiettivi di Eccellenza Scientifica di Horizon Europe e costituzione di reti” (CUP B53C22004310006).
- “EFC cod. SSU2024-00002, Missione 4 “Istruzione e ricerca” – Componente 1, “Potenziamento dell’offerta dei servizi all’istruzione: dagli asili nido all’università” – Investimento 3.4 “Didattica e competenze universitarie avanzate” – Sub-Investimento “Rafforzamento delle scuole universitarie superiori” (CUP: G97G24000100007).

Annex

Development environment

Both services were developed and tested in an isolated STENCIL virtual data-center on ORFEO before production deployment paths were considered.

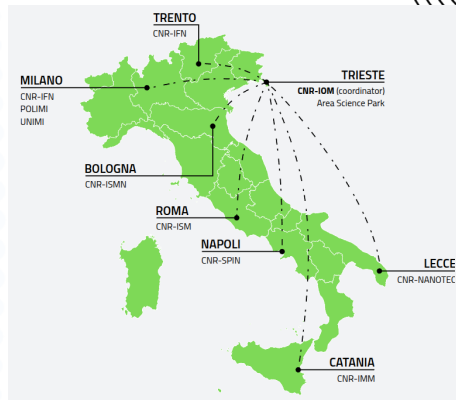
- **Boundary** one dedicated VM isolates development failures.
- **Storage** Ceph mirrors the S3-compatible substrate.
- **Identity** FreeIPA supplies internal DNS and auth services.
- **Runtime** k3s hosts both deployed thesis services.



NFFA-DI operational units codes

Local bucket names embed the institutional **UO code** (Unità Operativa), a machine-readable link between stored data and the producing institution.

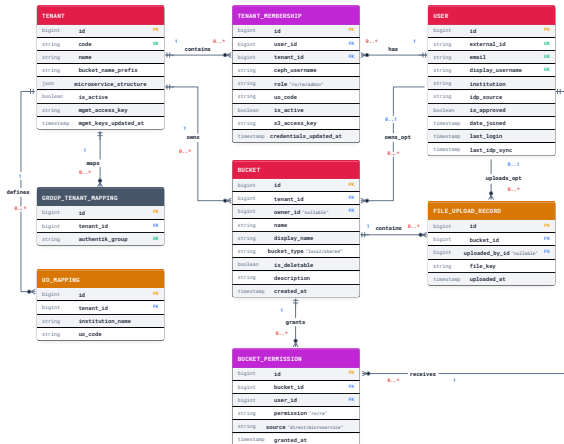
Institution	UO code
CNR – Officina dei Materiali, Trieste	cnr-iom.ts
CNR – Microelettronica e Microsistemi, Catania	cnr-imm.ct
CNR – Materiali Nanostrutturati, Bologna	cnr-ismn.bo
CNR – Fotonica e Nanotecnologie, Milano	cnr-ifn.mi
CNR – Fotonica e Nanotecnologie, Trento	cnr-ifn.tn
CNR – Nanotecnologia, Lecce	cnr-nanotec.le
CNR – Superconduttori (SPIN), Napoli	cnr-spin.na
CNR – Struttura della Materia, Roma	cnr-ism.rm
AREA – Ricerca e Innovazione Tecnologica	area-rit
POLIMI – POLIFAB, Milano	polifab
UNIMI – Fisica “Aldo Pontremoli”, Milano	unimi



Access model and data schema

Owner powers apply to local buckets only;
proposal-bucket roles come from upstream
proposal state.

Role	Create local	Delete local	Up-load	Down-load	Share local
Owner	✓	✓	✓	✓	✓
RW	—	—	✓	✓	—
RO	—	—	—	✓	—



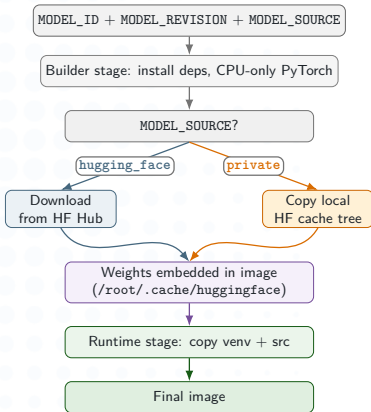
PostgreSQL schema: buckets, memberships, permissions, and upload records.

API contract and image build

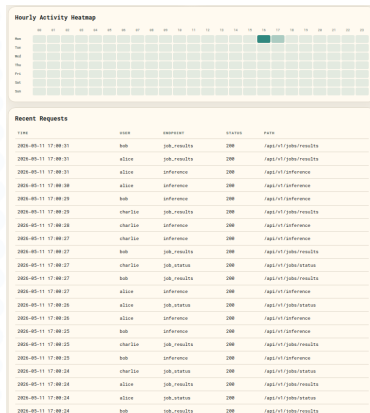
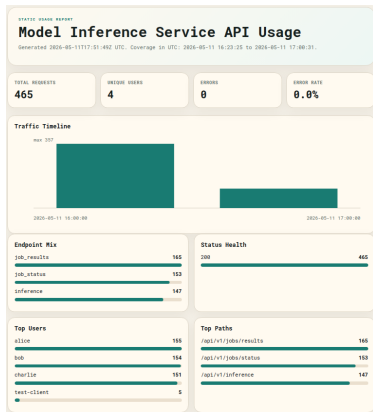
Public contract (POST)

Endpoint	Response
/api/v1/inference	job_id, status
/api/v1/jobs/status	state, timestamps
/api/v1/jobs/results	result, error, time

api_usage			
id	SERIAL		PK
timestamp	TIMESTAMPZ	NOT NULL	
username	VARCHAR(250)	NOT NULL	
service_name	VARCHAR(255)		
endpoint_type	VARCHAR(50)	NOT NULL	
status_code	INTEGER		
url_path	TEXT	NOT NULL	
Indexes: timestamp, username			



Operational reporting



Reports are generated from `api_usage`: per-user activity and endpoint mix over operator-selected windows.